

# Four Dials: Classroom Tasks on Explanation vs. Description

MYP Physics, companion task sheet to “Dials, Not Curtains”

A description tells you what happened once. An explanation tells you what would have happened if something else had been true, and pins that down precisely enough that a wrong guess about it would actually fail. In each of the four investigations below, you are looking for a **ratio**: a number built from the apparatus in front of you that controls whether some effect matters or can be ignored. Your job in each case is not just to notice that the effect exists, but to say exactly how it depends on the ratio, and to state a plausible-sounding wrong answer that your data could have caught you out on.

## 1. The spring

Hang a light spring vertically, unloaded, and look closely at the spacing between coils along its length.

**Task.** Sketch the coil spacing you observe, top to bottom. Where is it widest? Where is it narrowest? Suggest why, using the idea that each point in the spring only has to support the weight of the coils below it.

Now add a load to the bottom of the spring. Repeat with at least four different loads, from very light to as heavy as the spring can safely take, measuring the spacing of the top few coils and the bottom few coils each time.

*Prediction, before you run it:* As the load increases, will the difference between top-coil spacing and bottom-coil spacing get bigger, smaller, or stay the same? Write your prediction down before collecting data for the heavier loads.

*Rival worth ruling out:* One classmate says the unevenness should halve every time you double the load. Another says it should quarter every time you double the load. Which matches  $1 + 1/r$ , where  $r$  is load  $\div$  spring weight? Use your data to decide between them.

| Load | Top spacing | Bottom spacing | $r = \text{load} / \text{spring weight}$ |
|------|-------------|----------------|------------------------------------------|
|      |             |                |                                          |

**Write up:** state the ratio  $r$  you controlled, the relationship you predicted, and whether your data was precise enough to actually distinguish your prediction from the two rival guesses above.

## 2. The homemade thermometer

Build a large-bulb thermometer from a round-bottomed flask, a two-holed bung, and a length of narrow tube, filled completely with dyed water so the level sits part-way up the tube. Also get access to an ordinary small thermometer.

**Task.** Before you start timing, mark on your flask roughly where you expect the dye column to sit once the flask is fully warmed to the bath temperature.

Plunge both the homemade flask-thermometer and the small thermometer into the same warm water bath at the same moment. Record the small thermometer's reading (should be almost instant) and the dye column height in the flask at regular time intervals until it stops moving.

*Prediction, before you run it:* If you built a second flask with roughly double the radius of your first, how much longer do you expect it to take to settle: twice as long, four times as long, or eight times as long?

**Rival worth ruling out:** “Bigger bulb, slower response” is true but nearly unfalsifiable with one flask. The specific claim is that settling time scales with radius *squared*. If you or your group have built flasks of two different sizes, plot settling time against radius on your calculator or by hand on log-log paper and check whether the line has a gradient close to 2.

[illegible]

### 3. Bristles in oil

Set up a shallow dish of oil with cut hairs or fine bristles scattered in it, and two electrodes connected to a high-voltage supply at the edges of the dish (teacher-supervised).

**Task.** Before switching on, sketch what you expect the final bristle pattern to look like, based on the electrode shapes you are using.

Switch on and observe closely. Note whether any bristles overshoot their final orientation and wobble before settling, or whether they simply rotate smoothly into place.

*Prediction, before you run it:* If you repeat this in a noticeably more viscous oil, will the final pattern change, or only the time and manner of getting there?

*Rival worth ruling out:* “More viscous oil is just slower” misses a sharper claim: below some critical viscosity you should see genuine overshoot and wobble, and above it you should see none at all, not just a gradual slowing. If you have access to two oils of different thickness, try to find, roughly, where that switch happens.

## 4. The floating sphere

Drop a small sphere of known density into water, and record its depth over time using a phone camera against a ruled scale (a friend’s ruler taped to the side of the container works). Repeat with the same sphere dropped into a much more viscous fluid such as glycerol or golden syrup.

**Task.** Calculate, using the sphere’s density and the fluid’s density, the equilibrium depth you expect in each fluid *before* you drop it in.

*Prediction, before you run it:* Do you expect the equilibrium depth to be different in water compared to glycerol? Do you expect the *way it gets there* to be different?

*Rival worth ruling out:* This pairs directly with the bristle demonstration. Having found (or been told) whether  $\zeta < 1$  or  $\zeta \geq 1$  gives overshoot in that case, predict which regime each fluid will fall into here, and roughly how many bobs you expect in the underdamped case, before you look at the footage.

| Time (s) | Depth in water | Depth in glycerol | Notes |
|----------|----------------|-------------------|-------|
|          |                |                   |       |
|          |                |                   |       |
|          |                |                   |       |

## Reflection

- For each of the four investigations, name the ratio you controlled and the quantity it was built from.
- For *one* of the four, write out the specific, checkable relationship you tested, and the wrong-but-plausible alternative it ruled out.
- A good explanation should be *hard to vary*: changing its details should visibly break it. Pick one investigation and explain, in your own words, what would have had to be different about your data for your explanation to have been wrong.